SocialJev

A Family of Rapid Classification Foundation Models for the Social Sciences*

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First version: October 15, 2026 This version: October 25, 2026

Latest version: here

Abstract

We introduce SocialJev, a family of fast, performant, and scalable foundation models for classification in the social sciences. Built on Qwen3 transformers, the models replace autoregressive answer generation with a decision head that directly returns probabilities over arbitrary user-supplied classes. We fine-tune the models on classification queries constructed from 16 social-science datasets, covering topic presence, stance, sentiment, and framing. Test-set accuracy ranges from 96.1 to 98.0 percent across model sizes, with the smallest model processing approximately 310 Wikipedia articles per second on a single GH200 GPU.

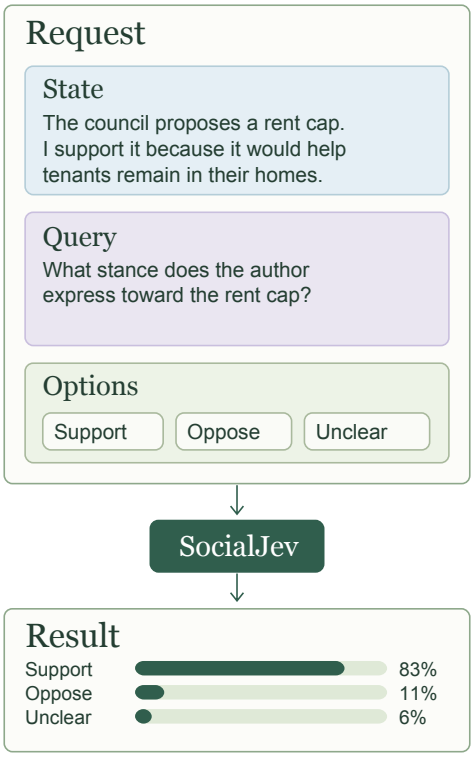
1 Introduction

Social scientists use text to measure political positions, public attitudes, institutional behaviour, and the framing of social problems (Grimmer and Stewart, 2013; Evans and Aceves, 2016; Gentzkow et al., 2019; Ash and Hansen, 2023). The relevant categories depend on the research question, often requiring distinctions defined in an application-specific codebook. Large language models (LLMs) can follow these instructions, but applying them to large corpora can be costly (Ash et al., 2026). Lightweight natural language processing methods are inexpensive to run, but typically require task-specific labelled data. Closed-source LLMs also create a reproducibility problem when providers change or retire models (Johnson and Obradovich, 2024; Palmer et al., 2024; Dunn et al., 2026; Wolfrath et al., 2026). Can small, open-weight models retain the flexibility of LLM classification while making annotation inexpensive and reproducible at scale?

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Model weights are available on Hugging Face, and the inference package is available on GitHub.

Figure 1: Example classification request and returned probabilities.



Notes: The example illustrates the model interface: a text (state), a classification query, and user-supplied answer options enter the model, which returns a probability for each option. The displayed probabilities are illustrative rather than evaluation results. Options can be specified at inference time and need not belong to a fixed label vocabulary.

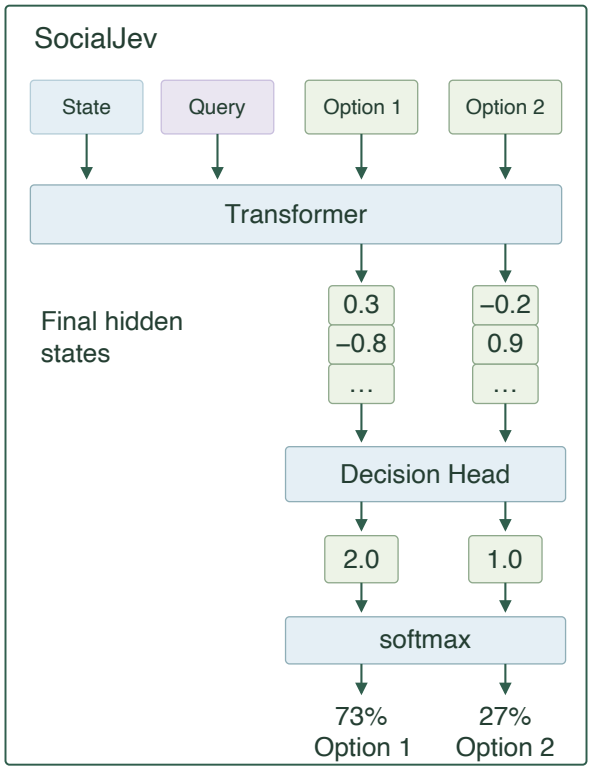
This paper makes progress by introducing SocialJev, a family of classification foundation models built on pretrained Qwen3 transformers. Researchers supply a text, a query, and answer options, and the model returns probabilities over those options. Figure 1 illustrates this interface. The categories can change between applications without requiring a new fixed output vocabulary. A decision head scores the supplied options directly, avoiding token-by-token answer generation. The approach is inspired by TypeSafe AI's Jev (Almeida, 2026), which demonstrates the promise of rapid decision models but remains closed-source.

We fine-tune SocialJev on classification queries constructed from 16 social-science datasets with human annotations and documented codebooks. The annotations guide synthetic query generation for topic presence, stance, sentiment, and framing, with variation in question wording and answer vocabularies. The architecture also preserves probabilities when queries or answer options are reordered, subject to floating-point differences. We release models with 0.6, 1.7, and 4 billion parameters.

Table 1 reports test-set accuracy ranging from 96.1 percent for the smallest model to 98.0 percent for the largest, which trails DeepSeek and GLM by 0.8 percentage points and Jev by 1.6 points. In a separate timing exercise on 100,000 sampled Wikipedia articles, the smallest model processes approximately 310 articles per second on one NVIDIA GH200. Extrapolating this rate to English Wikipedia gives approximately 6.5 hours at an estimated GPU rental cost of 15 US dollars, excluding preparation and startup. These exercises measure classification accuracy and processing speed separately, leaving accuracy on Wikipedia and broader transfer to new applications to further evaluation.

Open, versioned checkpoints and an accompanying inference package allow researchers

Figure 2: SocialJev architecture.



Notes: The pretrained Qwen3 backbone encodes the state (text), query, and answer options. A decision head maps each option's final hidden representation to a scalar score, and softmax normalises the scores into probabilities within the query. Released models use a linear head at 0.6B and 4B parameters and a multilayer perceptron with 512 hidden neurons at 1.7B. Diagram values are schematic, not estimated performance measures.

to retain the model used to construct their measures. SocialJev thus offers a reusable tool for large-scale text annotation while keeping application-specific validation central to its use in research.

2 Model

Option scoring. Jev-like models use standard causal transformers but score supplied answer options directly rather than generating responses token by token. Their *decision head* directly scores the probability of options supplied at inference time. TypeSafe's implementation is not public, but several open implementations have demonstrated comparable performance on public benchmarks. They differ primarily in how they retrieve scores from the backbone. Contrastive Language Models encode states and options separately, then score their similarity in a learned embedding space (Kwok et al., 2026). Liquid AI's d1 and Base Labs' Mammoth and Elephant repurpose the existing language-model head by appending letters to each option and evaluating the probability assigned to each letter (Liquid AI, 2026; Kirkby et al., 2026). Cloudflare's Clef instead uses a specialised decision head to score outputs permitted by the supplied schema (Chen et al., 2026).

We implement these core approaches in our setting and additionally test two simple decision heads. Figure 2 shows how these heads replace the language-model head of a pretrained Qwen3 transformer (Yang et al., 2025). The text (state), query, and candidate answer options enter the transformer, which produces a hidden representation for each option. A shared decision

head maps each option's final hidden representation to a scalar score. The linear head performs this mapping directly, while the multilayer perceptron (MLP) head uses one hidden layer and a nonlinear activation function. Finally, softmax normalises the scores into probabilities over the supplied options, as illustrated at the bottom of the figure.

We compare the architectures by held-out validation accuracy and their ability to generalise to query types excluded from training. In our setting, when jointly training the backbone and decision head, the linear and MLP variants perform best on both criteria at our current scale.

Order invariance. We consider order invariance a desirable property of decision models, particularly for scientific applications, where rearranging the order of queries or the options within a query should not change the probability assigned to each option. With conventional prompting, these properties are not guaranteed. LLMs exhibit positional biases, including preferences for particular answer identifiers, that can change their predictions when options are reordered (Zheng et al., 2024).

Following Set-Based Prompting (McIlroy-Young et al., 2024; Brown and McIlroy, 2025), we enforce these properties through the attention mask and positional encoding. Each query attends to the state but cannot attend to other queries. Each option attends to the state and its associated query but cannot attend to other options. Within each branch, attention remains causal. Figure 3 illustrates these restrictions.

The mask prevents information from passing between queries or competing options. We additionally assign position IDs relative to each branch's context, so reordering branches does not change their logical positions. Consequently, each option's score depends on the state, its query, and the option itself, rather than its placement relative to other options. Normalising these scores within each query yields probabilities invariant to query and option order, subject to floating-point differences.

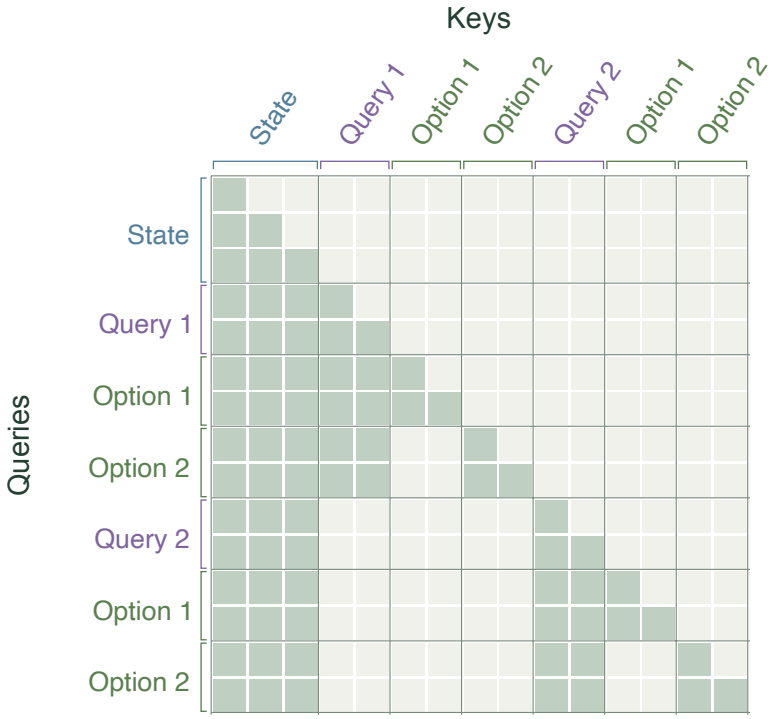
We use Qwen3 because its transformer architecture allows us to enforce this separation through attention masks and branch-relative position IDs. Newer hybrid models such as Qwen3.5 also mix information across tokens through recurrent layers and convolutions (Qwen Team, 2026), so attention masks alone are insufficient to impose the same restrictions.

3 Methods

We adapt the pretrained models to social-science classification by constructing queries from annotated texts and jointly fine-tuning the backbone and decision head. The training pipeline uses human annotations and codebooks to guide query generation, while attention masks and positional encodings enforce independence across queries and answer options. We describe the data construction, training objective, checkpoint selection, and test-evaluation setup below, followed by the Wikipedia sampling and timing protocol.

Datasets. We construct training states from 16 datasets sourced from the social sciences literature, covering political topics and positions, media bias, human values, environmental claims, and abusive language. We prioritise datasets with available texts, high-quality human annotations, and well-documented codebooks. The annotations and codebooks provide the basis for generating classification queries relevant to social-science research. Figure 4 shows the distribution of training queries across datasets.

Figure 3: Attention mask for states, queries, and options.



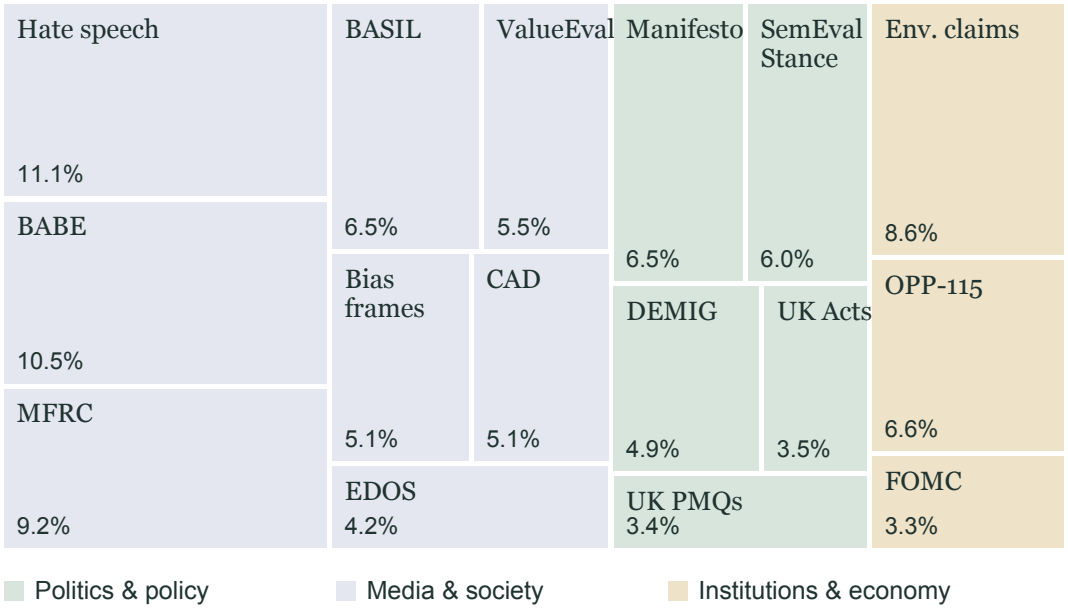
Notes: Rows indicate tokens receiving information and columns indicate tokens attended to. Filled cells permit attention, and axis colours identify the state (text), queries, and answer options. Each query attends to its state, and each option attends to its state and associated query. Attention between queries and between competing options is blocked, while attention within a branch remains causal. Branch-relative position IDs, described in the text, are also required for order invariance. When training examples are packed together, attention between different states is blocked.

Queries and Options. Using the states, human annotations, and codebooks as input, we generate training queries and options with Muse-Spark-1.3. The human annotations guide interpretation of the texts, while the codebooks define the concepts originally annotated. We generate queries for four core NLP tasks. Topic presence asks whether a text discusses a specified topic or concept. Stance asks whether it expresses support for or opposition to a specified target. Sentiment concerns a favourable, unfavourable, or mixed evaluation of a target, while framing concerns whether the text presents an issue through a specified interpretive frame.

Researchers differ in how they formulate these tasks, specify their coding instructions, and define the available options (e.g. Grimmer and Stewart, 2013). To improve robustness to this variation, the generation prompt encourages diverse wording, answer vocabularies, and formats, ranging from short questions to detailed codebook-style instructions.

Training procedure. We jointly fine-tune the pretrained transformer and newly initialised decision head using cross-entropy loss over the supplied answer options. The loss rewards assigning greater probability to the correct option, updating both the transformer representations and the head that converts them into scores. We optimise the transformer and head in separate parameter groups, allowing their learning rates to be chosen independently. For computational efficiency, we pack multiple examples into fixed-length sequences, reducing the space and computation spent on padding. The attention mask blocks information flow between examples associated with different texts, so packing does not allow one example to draw on another's context.

Figure 4: Composition of the training queries.



Notes: Rectangle areas represent each source dataset's share of training queries, and colours distinguish broad subject areas. Shares describe the training mixture rather than test-set performance or the prevalence of these subjects in social-science corpora. Sources are OPP-115 (Wilson et al., 2016); Moral Foundations Reddit Corpus (Trager et al., 2022); BABE (Spinde et al., 2021); Social Bias Frames (Sap et al., 2020); CAD (Vidgen et al., 2021); EDOS (Kirk et al., 2023); Environmental Claims (Stammbach et al., 2023); ValueEval (Kiesel et al., 2023); BASIL (Fan et al., 2019); UK Acts and UK PMQs (Bevan and Jennings, 2016; Bevan et al., 2011; Bevan, 2011; John et al., 2013); SemEval Stance (Mohammad et al., 2016); FOMC (Shah et al., 2023); DEMIG (DEMIG, 2015; de Haas et al., 2015); Manifesto Corpus (Lehmann et al., 2026); and Measuring Hate Speech (Sachdeva et al., 2022).

Model selection and evaluation. For each model size, we use held-out validation accuracy to choose the decision-head architecture and training checkpoint. The selected 0.6B and 4B models use linear heads, while the 1.7B model uses a multilayer perceptron with 512 hidden neurons. We then evaluate these selected checkpoints on a separate test set covering topic presence, stance, sentiment, and framing. We compare SocialJev with both specialised classification models and general-purpose language models. Jev, Clef, Liquid d1, and GLiNER provide comparisons with alternative approaches to rapid classification (Almeida, 2026; Chen et al., 2026; Liquid AI, 2026; Zaratiana et al., 2025), while DeepSeek and GLM provide reference points for classification using general-purpose LLMs (DeepSeek-AI, 2026; GLM-5 Team, 2026). Validation data thus guide model selection, while test results describe the performance of the selected models. The separate comparison of generalisation to query types excluded from training is discussed in Section 2.

Scaling Application with Wikipedia. To measure the throughput and cost of self-hosted annotation, we label 100,000 English Wikipedia articles randomly sampled from the October 2026 snapshot of Wikipedia Monthly (Kamali and Omneity Labs, 2025), using the inference package we supply for large-scale workloads with a vLLM backend.

We split long articles into passages of up to 4,096 tokens and ask one binary question per passage: "Does this passage discuss education policy?" All three models process the same material on one NVIDIA GH200, using the vLLM backend with 16 concurrent asynchronous requests, each containing a batch of 128 passages. Preparation and model startup are excluded from the run duration.

Table 1: Test-set classification accuracy overall and by task (percent).

Model	Overall	Topic	Stance	Sentiment	Framing
SocialJev 0.6B (preview)	96.1	98.8	93.5	96.2	94.6
SocialJev 1.7B (preview)	97.5	100.0	94.4	98.1	96.4
SocialJev 4B (preview)	98.0	100.0	96.8	95.3	99.1
TypeSafe Jev 1.13	99.6	100.0	100.0	98.1	100.0
DeepSeek V4.1 Flash	98.8	100.0	99.2	97.2	98.2
GLM 5.3 Flash	98.8	100.0	98.4	98.1	98.2
Clef	96.5	100.0	92.7	92.5	99.1
Clef Flash	93.4	99.4	88.7	90.6	92.0
Liquid d1-3B	90.0	98.8	85.5	82.1	89.3
GLiNER2.5-Decide	62.3	67.6	60.5	45.3	72.3
GLiNER2.5-multi-Decide	51.6	61.8	32.3	40.6	67.9

Notes: Entries are percentages of test queries classified correctly by the selected checkpoints and comparison models. Topic denotes topic or concept presence. Stance concerns support for or opposition to a target, sentiment concerns evaluation of a target, and framing concerns an interpretive frame. Overall is accuracy across the test set, rather than an unweighted average of the four task columns. SocialJev checkpoints are selected on validation data before test evaluation. The table reports point estimates without confidence intervals. Test-set counts and a chance benchmark are not reported here, so the percentages alone do not establish the precision of differences between models.

4 Results

We examine classification accuracy and inference throughput separately. The test-set comparison measures performance across the four classification tasks, while the Wikipedia exercise measures processing speed under a common hardware and inference configuration.

Classification performance. Table 1 reports overall and task-specific accuracy for the selected SocialJev checkpoints and comparison models. Overall accuracy ranges from 96.1 percent for the 0.6B model to 98.0 percent for the 4B model, with the 1.7B model attaining 97.5 percent.

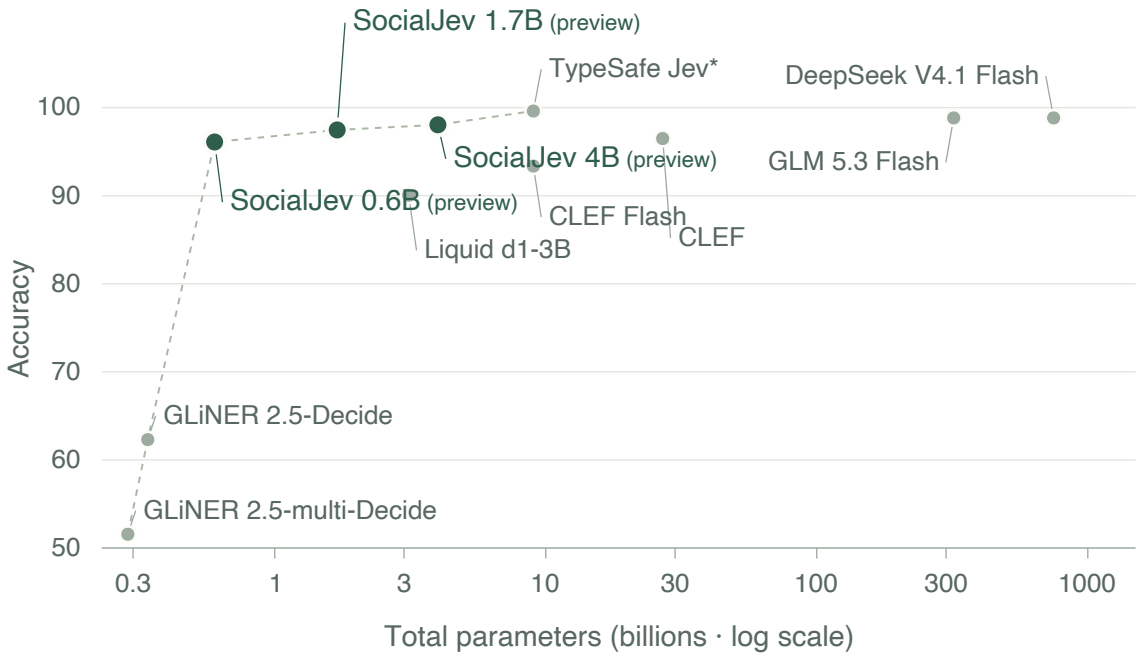
The task-specific results show that the relationship between model size and accuracy varies across classification problems. Stance accuracy rises from 93.5 percent for the 0.6B model to 96.8 percent for the 4B model, while framing accuracy increases from 94.6 to 99.1 percent. Topic accuracy is already 98.8 percent for the smallest model and reaches 100.0 percent for the two larger models. Sentiment accuracy is not monotonic in model size: the 1.7B model scores 98.1 percent, compared with 96.2 percent for the 0.6B model and 95.3 percent for the 4B model.

Figure 5 places these results alongside model size. Overall accuracy increases from 96.1 percent for SocialJev 0.6B to 98.0 percent for SocialJev 4B. The latter trails DeepSeek and GLM by 0.8 percentage points and Jev by 1.6 points. The preview models therefore approach the accuracy of the strongest comparison models on this test set while using compact, openly available checkpoints. Jev's parameter count is not disclosed, so its location in the figure relies on an external estimate.[1]

Classification speed and cost. To assess the practicality of classifying large text collections, we measure how quickly each model processes the same sample of Wikipedia articles on a

[1] JevResearch estimates 4–9 billion parameters under a dense-model interpretation (JevResearch, 2026). The figure uses the upper end of this range.

Figure 5: Test-set classification accuracy and model size.



Notes: Accuracy is the overall percentage of correctly classified test queries, as reported in Table 1. Green points identify SocialJev models, and the dashed line marks the plotted Pareto frontier in accuracy and parameter count. Jev is plotted at 9 billion parameters, the upper end of JevResearch's estimated 4–9 billion range under a dense-model interpretation (JevResearch, 2026), rather than a disclosed model size. The figure shows point estimates and does not report confidence intervals.

single GPU. Figure 6 reports articles processed per second in the left panel and input tokens processed per second over the run in the right panel. SocialJev 0.6B processes approximately 310 articles per second, compared with 160 for SocialJev 1.7B and 68 for SocialJev 4B. The smallest model is therefore approximately twice as fast as the 1.7B model and 4.6 times as fast as the 4B model under this configuration.

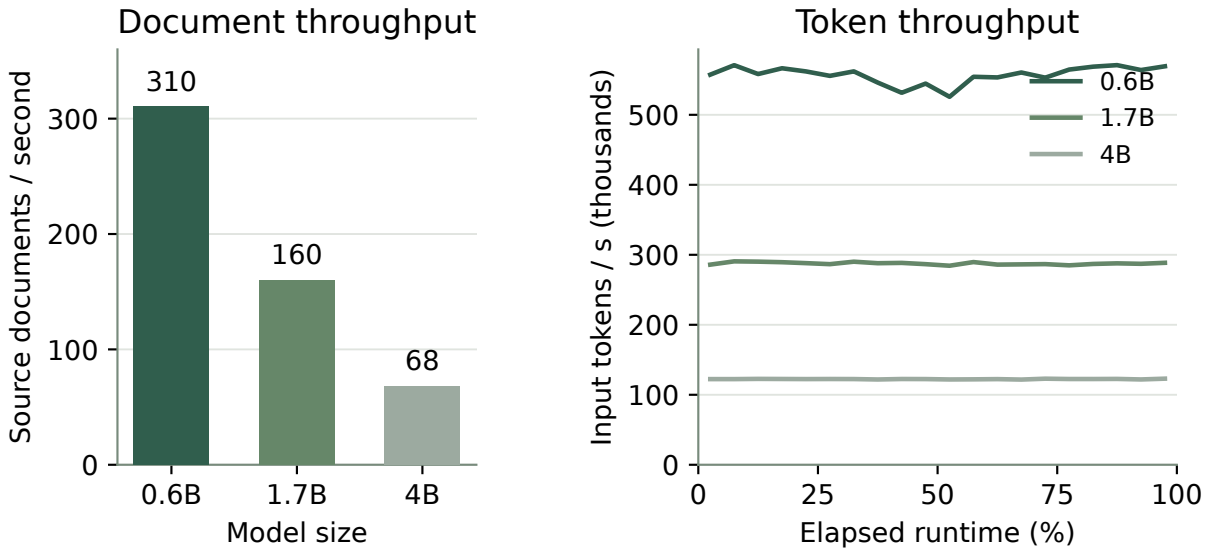
These speed differences help put the accuracy results in practical terms. Moving from the smallest to the largest model raises overall accuracy on classification at the cost of lower speed. Extrapolating the 0.6B run to all English Wikipedia gives approximately 6.5 hours on a single NVIDIA GH200, at an estimated GPU rental cost of 15 US dollars. Additional questions, longer coding instructions, or different hardware may change processing time and cost.

5 Conclusion

We introduce SocialJev to combine the flexibility of LLMs with the speed and low cost of traditional NLP methods, retaining reproducibility through open, versioned weights. SocialJev bypasses autoregressive decoding and returns probabilities over arbitrary options directly, rather than requiring fixed output classes and dimensions. Fine-tuning across social-science tasks is intended to make the models useful without further task-specific adaptation.

The preview models approach the accuracy of substantially larger models while remaining small enough to host inexpensively on university or rented hardware. Together, these properties make large-scale transformer-based annotation accessible without dependence on a closed-weight provider's continued service. We release three model sizes and an inference package. Future development will expand the training data and scale the models, add extraction with

Figure 6: Wikipedia inference throughput on one NVIDIA GH200.



Notes: The workload comprises 100,000 English Wikipedia articles randomly sampled from the October 2026 Wikipedia Monthly snapshot. Articles are split into passages of at most 4,096 tokens and classified using the binary query about education policy stated in the text. All three models process the same material using vLLM, with 16 concurrent asynchronous requests and 128 passages per request. The left panel reports articles per second. The right panel reports input tokens per second, including cached prefixes, over normalised elapsed runtime, with rates measured in five-percent intervals. Preparation and model startup are excluded. These are throughput measurements, not measures of classification accuracy.

citations to supporting text and simple numerical features, and extend the models to multimodal inputs. We also plan to integrate GPU providers to simplify deployment and provide interfaces for fine-tuning, evaluation, and query optimisation.

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